Prefix Sum — L and R Range Flow

This note focuses only on one thing:

> How `L` and `R` from the raw/original array are used with the two common Prefix Sum conventions.

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1. Starting Point — RAW ARRAY

We always start with the original array:

```
arr = [2, 4, 6, 8, 10]

index:  0  1  2  3   4
value:  2  4  6  8  10
```

Suppose we want the range:

```
L = 1
R = 3
```

So we want:

```
arr[1] + arr[2] + arr[3]

= 4 + 6 + 8

= 18
```

Important:

`L` and `R` come from the **raw/original array**.

They are the boundaries of the range we want.

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2. Concept 1 — Prefix Sum INCLUDING `i`

Here, the Prefix Sum array stores the sum **including the current index**.

Step 1 — RAW ARRAY

```
arr = [2, 4, 6, 8, 10]

index:  0  1  2  3   4
```

Step 2 — Choose the range

```
L = 1
R = 3
```

So:

```
4 + 6 + 8 = 18
```

Step 3 — Build PREFIX ARRAY

```
prefix = [2, 6, 12, 20, 30]

index:    0   1   2   3   4
prefix:   2   6  12  20  30
```

Here:

```
prefix[i] = arr[0] + ... + arr[i]
```

Step 4 — Use `L` and `R`

For this convention:

```
sum(L, R) = prefix[R] - prefix[L - 1]
```

So:

```
L = 1
R = 3
```

becomes:

```
prefix[3] - prefix[0]
```

Step 5 — Calculate

```
prefix[3] - prefix[0]

= 20 - 2

= 18
```

Which gives:

```
4 + 6 + 8 = 18
```

Why?

```
prefix[3] = 2 + 4 + 6 + 8
prefix[0] = 2
```

Subtract:

```
(2 + 4 + 6 + 8) - 2

= 4 + 6 + 8

= 18
```

IMPORTANT

Because this Prefix Sum includes `i`, we remove everything before `L` using:

```
L - 1
```

So:

```
sum(L, R) = prefix[R] - prefix[L - 1]
```

For `L = 0`, this convention needs a separate edge case unless the prefix array is built with an extra zero.

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3. Concept 2 — Prefix Sum EXCLUDING `i`

This is the **extra `0` convention**.

Here:

```
prefix[i] = sum of elements BEFORE index i
```

Step 1 — RAW ARRAY

```
arr = [2, 4, 6, 8, 10]

index:  0  1  2  3   4
```

Step 2 — Choose the range

```
L = 1
R = 3
```

So again:

```
4 + 6 + 8 = 18
```

Step 3 — Build PREFIX ARRAY

```
prefix = [0, 2, 6, 12, 20, 30]

index:    0  1  2   3   4   5
prefix:   0  2  6  12  20  30
```

Here:

```
prefix[i] = sum of elements before i
```

For example:

```
prefix[1] = 2
prefix[2] = 2 + 4 = 6
prefix[3] = 2 + 4 + 6 = 12
prefix[4] = 2 + 4 + 6 + 8 = 20
```

Step 4 — Use `L` and `R`

For this convention:

```
sum(L, R) = prefix[R + 1] - prefix[L]
```

So:

```
L = 1
R = 3
```

becomes:

```
prefix[4] - prefix[1]
```

Step 5 — Calculate

```
prefix[4] - prefix[1]

= 20 - 2

= 18
```

Which gives:

```
4 + 6 + 8 = 18
```

Why?

```
prefix[4] = 2 + 4 + 6 + 8
prefix[1] = 2
```

Subtract:

```
(2 + 4 + 6 + 8) - 2

= 4 + 6 + 8

= 18
```

IMPORTANT

Because this Prefix Sum stores the sum **before** `i`, we use:

```
R + 1
```

to make sure `arr[R]` is included.

So:

```
sum(L, R) = prefix[R + 1] - prefix[L]
```

---

4. The Two Flows Side by Side

Both start with the **same raw array** and the **same `L` and `R`**.

```
RAW ARRAY

arr = [2, 4, 6, 8, 10]
        0  1  2  3   4

L = 1
R = 3
```

We want:

```
4 + 6 + 8 = 18
```

INCLUDING `i`

Build:

```
prefix = [2, 6, 12, 20, 30]
```

Formula:

```
prefix[R] - prefix[L - 1]
```

Put in the values:

```
prefix[3] - prefix[0]

= 20 - 2

= 18
```

EXCLUDING `i`

Build:

```
prefix = [0, 2, 6, 12, 20, 30]
```

Formula:

```
prefix[R + 1] - prefix[L]
```

Put in the values:

```
prefix[4] - prefix[1]

= 20 - 2

= 18
```

---

5. The Main Difference

The **raw array range is the same**:

```
L = 1
R = 3
```

Only the Prefix Sum convention changes.

Including `i`

```
Prefix meaning:
prefix[i] = sum from 0 to i

Range:
sum(L, R) = prefix[R] - prefix[L - 1]
```

Excluding `i`

```
Prefix meaning:
prefix[i] = sum before i

Range:
sum(L, R) = prefix[R + 1] - prefix[L]
```

---

6. The Most Important Mental Model

Always think in this order:

```
1. RAW ARRAY
       ↓
2. Choose L and R
       ↓
3. Build PREFIX ARRAY
       ↓
4. Use L and R in the correct formula
       ↓
5. Get the range sum
```

Do **not** think that `L` and `R` come from the Prefix Sum array.

They describe the range in the **original/raw array**.

The Prefix Sum array is only a helper used to calculate the answer quickly.

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7. Quick Memory Trick

Including `i`

```
prefix includes the current index
```

Therefore:

```
sum(L, R)

= prefix[R]
- prefix[L - 1]
```

Think:

> Go to `R`, then remove everything before `L`.

Excluding `i`

```
prefix stores what was BEFORE the current index
```

Therefore:

```
sum(L, R)

= prefix[R + 1]
- prefix[L]
```

Think:

> Go one step past `R` so that `R` is included.

---

8. Final Example

```
arr = [2, 4, 6, 8, 10]

L = 1
R = 3
```

Target range:

```
[4, 6, 8]

sum = 18
```

Including `i`

```
prefix = [2, 6, 12, 20, 30]

prefix[R] - prefix[L-1]

prefix[3] - prefix[0]

20 - 2

= 18
```

Excluding `i`

```
prefix = [0, 2, 6, 12, 20, 30]

prefix[R+1] - prefix[L]

prefix[4] - prefix[1]

20 - 2

= 18
```

Same answer.

Only the **Prefix Sum definition and formula** are different.